

Distance Measurement System using Arduino UNO

Abstract

This project measures distance using HC-SR04 and Arduino UNO.

Objectives

Measure distance accurately, learn sensor interfacing and Arduino programming.

Components

Arduino UNO, HC-SR04, Breadboard, Jumper Wires, USB Cable.

Working Principle

The Arduino sends a trigger pulse. The sensor returns an echo. The echo time is converted into distance using the speed of sound.

Applications

Obstacle detection, robotics, parking assistance, automation.

Advantages

Low cost, easy to build, real-time measurement.

Conclusion

A simple embedded system demonstrating ultrasonic distance measurement.